

CSCI 566: Deep Learning and its Applications

Fall 2026 • Mondays • 5:00pm-8:20pm (4 units)

Section	Format	Location	Capacity
30122	Lecture, in person	SGM 124	231
30359	Lecture, DEN@Viterbi	Remote	30

Both sections meet at the same time and receive the same lecture. Where a policy differs between sections, this syllabus says so explicitly.

- Course website: <https://usc-fortis.github.io/CSCI566-F26/>
- Course Piazza: to be announced in week 1
- Course Google Drive (slides): to be announced in week 1

Course Description

This course offers a comprehensive introduction to the principles of machine learning (ML) and deep learning (DL), emphasizing both mathematical foundations and practical applications. You will gain insights into basic ML techniques, learn advanced DL applications in fields such as computer vision and natural language processing, and understand their impact on areas such as image recognition and autonomous systems. The course includes a hands-on assignment and a customizable final project, giving you practical experience in implementing ML and DL solutions.

Guest lectures. Industry and academic professionals join our lectures regularly, sharing their experience in ML and data science and providing career insights.

Prerequisites

- Proficiency in Python
- College calculus, linear algebra
- Probability and statistics
- Recommended: CSCI 567 (Machine Learning) or equivalent

Recommended preparation: sufficient mathematical background, good programming skills, and familiarity with concepts and methods in machine learning and AI.

Duplicate credit: CSCI 527.

Instructor and Teaching Assistants

Instructor: Yue Zhao

Office hours: During and after the Monday lecture block. DEN students may join the same office hours remotely; see the course website for the link.

Office hours are for non-homework and non-project inquiries. Questions about course content and logistics are encouraged. Direct technical questions to the TAs.

Teaching assistants: Three teaching assistants support this course. Their names, office hours, and contact details are on the course website. TA sessions focus on homework, project guidance, and administrative support.

Contact: Direct all course communication through Piazza. Given the size of the course (261 students across two sections), we prioritize Piazza over email.

Required Readings and Supplementary Materials

- *Deep Learning* (MIT Press) by Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Free online: <http://www.deeplearningbook.org/>
- *Mathematics for Machine Learning* (Cambridge University Press) by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong. Free online: <https://mml-book.github.io/>

Description and Assessment of Assignments

Two in-class quizzes at announced times, one unannounced in-class quiz, one coding assignment, one midterm exam, and one course project (pre-proposal, reports, presentations).

There is no final exam. The final project report and presentation serve as the final summative experience for this course and are due during the university's final examination period.

Grading Breakdown

Deliverable	Points
In-class quizzes	9 (Quiz 1: 5, Quiz 2: 4)
Unannounced quiz	3
Assignment	10 (one)
Midterm	40
Course project	38
Total	100

Deliverable	Points

In-Class Quizzes

Quizzes are graded on a completion basis, and answers are reviewed together during the same class session. Quizzes must be completed within the designated class period. There are no makeup quizzes.

Both sections take quizzes during the live session. DEN students complete the quiz in the same window as the in-person section.

Assignment

One individual coding task. Submissions include both code and documentation. You will be given starter code, a description of the required implementation, and unit tests that check compatibility with the grading scripts.

Midterm Exam

Format: Open book, no electronic devices. Mostly multiple-choice questions plus some computation-based questions with unambiguous single-number answers.

Date: Monday, November 2, in the regular class session.

Content: Everything covered in class up to and including the lecture on October 26, the session before the exam. **If a topic is not reached by that date, it is not on the exam.** The scope follows what was actually taught, so there is no separate module list to track.

Detailed scope. A specific exam scope, listing which topics are and are not tested, is posted on the course website and Piazza by **October 19**, two weeks before the exam. That posted scope is always a **subset** of the material above: it can narrow what is tested, never add to it. Study from the posted scope once it is available.

A review session is held in class on **October 26**, the week before the exam.

DEN section: Students in section 30359 take the exam on the same day. The exact delivery and proctoring arrangement is confirmed with DEN@Viterbi and announced on the course website before the exam.

Makeup exams are not provided unless Viterbi approves the supporting documents.

Academic Integrity on the Midterm

Any academic integrity violation on the midterm results in a failing grade for the course. This is not a deduction on the exam and not a zero on that component. It is an F for CSCI 566, and the case is reported to the university under SCampus Section 11.

This applies to, and is not limited to: copying from another student, allowing another student to copy

from you, using any electronic device during the exam, using unauthorized materials, and communicating with anyone other than course staff during the exam period.

We use multiple exam versions and assigned seating. Proctors record the seat assignment for every student.

This policy exists because the exam is worth 40 points and is taken by 261 students. A weaker penalty transfers the cost of cheating onto the students who do not cheat.

Course Project

Groups of 4 to 6 students. The topic is open, ranging from applied ML projects to rigorous ML research projects. This is an applied course, so publication-track projects are rare.

Each group is assigned a TA, who meets with the group twice to check that the project is progressing.

Project matching. A form collects your project preferences and potential teammates. Piazza supports this process.

Groups may mix the two sections, and doing so is encouraged. A group with both in-person and DEN members presents at the poster session through its in-person members.

Component	Share of project grade
Pre-proposal (1 page: team, initial thoughts, planned dataset)	10%
Midterm report (progress, literature review, initial results, updated plan)	30%
Final report and presentation	60%

The final report should resemble a research paper: motivate the problem, present the approach, describe the experiments, and discuss the results.

Final presentation format. The last class (November 30) is a **poster and demo session**, not sequential talks. All groups present in parallel; the instructor and the TAs visit each poster for about five minutes, and **every group member answers questions individually**, including about parts of the project they did not personally write.

The reason is arithmetic: 261 students in groups of five is roughly 53 groups, and a single 200-minute session gives under four minutes per group if presentations run one after another. A parallel session removes that limit and makes real questions possible.

Peer Evaluation Within the Group

Part of the final report and presentation grade is individual and is determined by peer evaluation within your group.

How it works.

1. After the final presentation, every group member scores every other member of the same group on agreed contribution criteria.
2. Your individual peer score is the aggregate of the scores your teammates gave you.
3. **You see your own aggregate score. You do not see who gave which score.**
Individual ratings are visible only to course staff.
4. Submitting a peer evaluation is required. Failing to submit forfeits your own peer score.

Weight. A portion of the 60% final report and presentation component is set by peer evaluation, and the remainder is the shared group grade. The exact share is announced at the start of the term.

Why this exists. In previous offerings, every group member received the same project grade regardless of contribution, and course evaluations reported that this left groups with no mechanism to address uneven participation. Peer evaluation gives that mechanism while keeping most of the grade collective, so the group still has a shared incentive to produce good work.

Course staff review peer scores before they are applied. If the pattern within a group suggests retaliation or collusion rather than a contribution difference, staff may adjust or discard the peer component for that group.

Assignment Submission Policy

All assignments and project reports are due in electronic form by 11:59pm Pacific on the due date. There are no late days.

Grading Scale

Grade	Range	Grade	Range
A	93-100	C+	77-79
A-	90-92	C	73-76
B+	87-89	C-	70-72
B	83-86	D+	67-69
B-	80-82	D	63-66
		D-	60-62
		F	59 and below

Letter grades are decided by rounding floating-point grades up to the nearest whole number (for example 92.2 becomes A, and 59.8 becomes D-).

Schedule

The course meets on 14 Mondays. September 7 is Labor Day and the university is closed. The last class is November 30; December 7 falls in the study and examination period. The most current schedule is on the course website.

#	Date	Topic	Assessment
1	Aug 24	Course intro; ML and DL framing; agent security and trustworthy AI; linear models	
2	Aug 31	Classical ML: decision trees, kNN, clustering; model metrics (recorded lecture, no live session)	
—	Sep 7	Labor Day, no class	
3	Sep 14	Neural network basics: perceptron, gradient descent, forward propagation	
4	Sep 21	Backpropagation; vanishing gradients; activation functions; NVIDIA guest lecture	Assignment out
5	Sep 28	Convolutional neural networks	Quiz 1
6	Oct 5	RNN and LSTM; graph neural networks	Pre-proposal due
7	Oct 12	Training dynamics; transfer learning	
8	Oct 19	Generative AI: GAN and VAE	Assignment due; exam scope posted
9	Oct 26	Attention, decoding, mixture of experts; midterm review	
10	Nov 2	Midterm	Midterm
11	Nov 9	Self-supervised learning and LLMs; out-of-distribution and anomaly detection	Project midterm report due
12	Nov 16	Scaling laws; reinforcement learning	Quiz 2
13	Nov 23	Automated ML; project clinic	
14	Nov 30	Project poster session	
—	Final exam	No final exam	Final report and

#	Date	Topic	Assessment
	period		presentation due

The schedule is subject to change. The most current version is on the course website.

Statement on Academic Conduct and Support Systems

Academic Conduct

Plagiarism, presenting someone else's ideas as your own either verbatim or recast in your own words, is a serious academic offense with serious consequences. Please familiarize yourself with the discussion of plagiarism in SCampus Section 11, Behavior Violating University Standards: <https://scampus.usc.edu/1100-behavior-violating-university-standards-and-appropriate-sanctions>

Other forms of academic dishonesty are equally unacceptable. See additional information in university policies on scientific misconduct: <http://policy.usc.edu/scientific-misconduct>

The midterm integrity policy stated above is in addition to, not in place of, these university policies.

Discrimination, sexual assault, and harassment are not tolerated by the university. You are encouraged to report any incidents to the Office of Equity and Diversity <http://equity.usc.edu> or to the Department of Public Safety

<http://capsnet.usc.edu/department/departement-public-safety/online-forms/contact-us> This is important for the safety of the whole USC community. Another member of the university community, such as a friend, classmate, advisor, or faculty member, can help initiate the report or can initiate it on behalf of another person. The Center for Women and Men

<http://www.usc.edu/student-affairs/cwm/> provides 24/7 confidential support, and the sexual assault resource center webpage <http://sarc.usc.edu> describes reporting options and other resources.

Support Systems

A number of USC's schools provide support for students who need help with scholarly writing. Check with your advisor or program staff to find out more. Students whose primary language is not English should check with the American Language Institute <http://dornsife.usc.edu/ali>, which sponsors courses and workshops specifically for international graduate students. The Office of Disability Services and Programs http://sait.usc.edu/academicsupport/centerprograms/dsp/home_index.html provides certification for students with disabilities and helps arrange the relevant accommodations. If an officially declared emergency makes travel to campus infeasible, USC Emergency Information <http://emergency.usc.edu> will provide safety and other updates, including ways in which instruction will be continued.